

Order Cancellation Driver Analysis

Business Analyst Case Study | Dataset: Olist E-Commerce (seller marketplace)

00 · EXECUTIVE SUMMARY

Executive Summary

This report investigates order cancellations and unfulfilled orders across a 99,441-order e-commerce marketplace dataset. Rather than treating a low aggregate cancellation rate (1.24%) as negligible, the analysis segments the data to test whether the problem is concentrated in a specific, actionable part of the business.

The analysis finds that **seller tenure is a statistically validated driver of cancellation risk**: sellers active under 30 days cancel at 1.08%, versus 0.40% for sellers active 90+ days — a gap confirmed via chi-square testing ($p < 0.001$) to be real, not random variation. This pattern holds even within individual product categories, ruling out confounding.

The recommendation is a tenure-based onboarding / quality gate for new sellers, prioritized by category impact, along with a standing pod-level metric to monitor the new-seller cohort going forward.

Report Structure

This analysis is documented as a set of focused, standalone sections rather than one long document, so each part can be reviewed independently:

Doc #	Section	Contents
01	Problem Statement	Business context, why the metric matters, scope and data source
02	Hypotheses & Examples	The 3 hypotheses tested, with worked numeric examples
03	SQL Methodology & Queries	Full query logic behind every segmentation cut
04	Statistical Validation	Chi-square test, calculation walkthrough, interpretation
05	Findings & Priority Segments	Segment-level results and the priority ranking
06	Recommendations	Actions, prioritization logic, and success metrics

Key Numbers at a Glance

Metric	Value
Total orders analyzed	99,441
Overall cancellation / unavailable rate	1.24%
New-seller (0-30 days) cancellation rate	1.08%
Established-seller (90+ days) cancellation rate	0.40%
Chi-square statistic (tenure vs. cancellation)	102.83

Metric	Value
Statistical significance	$p < 0.001$
Highest-impact segment	Automotive category, new sellers (2.51% rate)

Data Source & Scope

This analysis uses the public Olist Brazilian E-Commerce dataset as a proxy for a multi-seller marketplace. It was chosen because its structure (orders, sellers, products, order items) closely mirrors a seller-marketplace model. The methodology — SQL-based segmentation, hypothesis testing, statistical validation, and priority-ranked recommendations — is directly transferable to a live marketplace dataset such as one used internally at a company like Meesho.